

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

To strengthen the organization's security and help prevent another data breach, a few key protections should be put in place.

First, the company should turn on **multi-factor authentication (MFA)** for all employee accounts. This helps because even if someone gets hold of a password, they still can't log in without a second step like a code from a phone or an authentication app. It adds an extra layer of protection that makes unauthorized access much harder.

Second, the organization needs to fix its password practices by requiring **strong, unique passwords and removing any default credentials**, especially for the database admin account. Right now, using a default password is a serious risk because attackers often already know those defaults. Strong passwords and regular updates make it much harder for attackers to guess or reuse credentials.

Lastly, the company should set up and properly configure **firewall rules to control network traffic**. This means only allowing trusted traffic in and out of the system and blocking anything suspicious. Without these rules, attackers have too much freedom to move through the network or send data out without being stopped.

Together, these steps would significantly reduce the chances of another breach and make the network much more secure.

Part 2: Explain your recommendations

To address the vulnerabilities found in the organization's network, the recommended security measures work together to reduce risk and strengthen overall protection.

Enabling **multi-factor authentication (MFA)** is important because it adds an extra layer of security beyond just passwords. Even if an attacker steals or guesses a password, they still cannot access an account without completing a

second verification step. This greatly reduces the risk of unauthorized logins, especially for employee accounts that may be targeted through phishing or credential theft.

Improving **password security by removing default credentials and enforcing strong passwords** is another critical step. Default admin passwords are widely known and often targeted first by attackers, making them an easy entry point into sensitive systems like databases. Replacing them with strong, unique passwords and requiring regular updates makes it much harder for attackers to gain access through guessing or reused credentials.

Finally, implementing **proper firewall rules** helps control what traffic is allowed into and out of the network. Firewalls act as a barrier between trusted internal systems and external networks. By filtering traffic and blocking suspicious connections, the organization can prevent many types of attacks and reduce the chance of data being stolen or systems being exploited.

Overall, these recommendations improve security by strengthening authentication, protecting sensitive accounts, and controlling network access.

Summary: One of the best ways to prevent this kind of issue from happening again is to use **multi-factor authentication (MFA)**. It works well because it adds an extra step when logging in, not just a password. So even if someone manages to steal or guess a password, they still can't get in without that second verification step, like a code sent to a phone or an authentication app. This makes it much harder for attackers to break into accounts.

MFA should always be turned on and used all the time for all employees, especially for admin accounts and anything that has access to important data.

It's not something you set once and forget—you need to keep it enabled at all

times and make sure it doesn't turn off. It should also be checked regularly to make sure everyone is still using it properly.